CCES – College of Computer Engineering and Sciences

CS3401 – Algorithm Design and Analysis

Chapter 6- Transform & Conquer
Linear Programming

References:

- *Introduction to the Design and Analysis of Algorithms. Anany V. Levitin, Pearson 3rd Edition.*

Transform & Conquer

Problem reduction –Linear Programming

- A ***Linear Programming (LP) model*** is a mathematical model that represents an optimization problem (minimization or maximization) where both the objective function and the constraints are linear.

- The basic components of an LP model include:
  - *Data (or Parameters)*
  - *Decision Variables*
  - *Constraints*
  - *Objective Function(s)*

- LP model is a commonly used model in mathematical programming.
  - Constraints may be equalities or inequalities.
  - Decision variables are non-negative and can be continuous and/or discrete (integers or binary)

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Problem reduction –Linear Programming

- Many problems of optimal decision making can be reduced to an instance of the ***linear programming*** problem.
- ***Linear programming*** problem: problem of optimizing a linear function of several variables subject to constraints in the form of linear equations and linear inequalities.

Objective Function

maximize (or minimize) $c_1 x_1 + \ldots + c_n x_n$

Decision Variables

subject to

Constraints

$a_{i1} x_1 + \ldots + a_{in} x_n \leq$ (or $\geq$ or $=$) b_i for $i = 1, \ldots, m$

$x_1 \geq 0, \ldots, x_n \geq 0$ **Non-negativity restriction**

LP Model Characteristics

- The Linear Programming (LP) model has the following key characteristics:

  1. ***Objective Function***
     - An LP model must have a single objective function that needs to be maximized or minimized.
     - The objective function is expressed as a linear combination of the decision variables.

  2. ***Decision Variables***
     - An LP model must have one or more decision variables that represent the choices to be made.
     - The decision variables can be continuous (taking any value within a range) or discrete (taking specific values, such as integers).

  3. ***Constraints***
     - An LP model must have one or more constraints that define the feasible region.
     - The constraints are expressed as linear equations or inequalities involving the decision variables.

  4. ***Linearity***
     - Both the objective function and the constraints must be linear, meaning that they can only involve addition, subtraction, and multiplication by constants.
     - No non-linear terms (e.g., products of decision variables, exponents, logarithms) are allowed.

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LP Model Characteristics

- The Linear Programming (LP) model has the following key characteristics:

1. ***Non-Negativity***
   - In many LP models, the decision variables are required to be non-negative (i.e., greater than or equal to zero).
   - This reflects real-world scenarios where negative values may not make sense (e.g., quantities of products, resources).

2. ***Deterministic Nature***
   - **LP models** are typically **deterministic**, meaning that all parameters (coefficients in the objective function and constraints) are known with certainty.
   - There is no consideration of uncertainty or randomness in the model.

3. ***Finiteness***
   - An **LP model** has a **finite number** of **decision variables** and **constraints**, making it computationally manageable.
   - This allows for the application of various **solution techniques** to find the optimal solution efficiently.

4. ***Linearity***
   - Both the objective function and the constraints must be linear, meaning that they can only involve addition, subtraction, and multiplication by constants.
   - No non-linear terms (e.g., products of decision variables, exponents, logarithms) are allowed.

A Simple Example

Problem statement: A manufacturing company produces two products, $Prod_1$ and $Prod_2$, using two raw materials, M_1 and M_2. The company aims to determine the optimal production quantities (number of tons) of each product to maximize its profit while adhering to the availability constraints of the raw materials.

The following data is provided:

- Each unit of $Prod_1$ requires 6 units of M_1 and 1 unit of M_2.
- Each unit of $Prod_2$ requires 4 units of M_1 and 2 units of M_2.
- The profit obtained from selling one unit of $Prod_1$ is \$5, while the profit from selling one unit of $Prod_2$ is \$4.
- The available quantities of raw materials are 24 units of M_1 and 6 units of M_2.

Data summary:

Raw Material	Usage for $Prod_1$	Usage for $Prod_2$	*Material Availability*
M_1	6	4	**24**
M_2	1	2	**6**
Profit	**\$5**	**\$4**	

A Simple Example

To formulate the **LP model** for this problem, **we need to define the following components**:

- **Decision variables:**
  - Let X_1 be the number of units produced of $Prod_1$.
  - Let X_2 be the number of units produced of $Prod_2$.
- **Objective function:** Maximize the total profit Z from producing $Prod_1$ and $Prod_2$, given by $\mathbf{Max} Z = 5X_1 + 4X_2$
- **Constraints:**
  1. M_1 availability constraint: $6X_1 + 4X_2 \leq 24$
  2. M_2 availability constraint: $1X_1 + 2X_2 \leq 6$
  3. Non-negativity constraints: $X_1, X_2 \geq 0$

Remarks

- The provided **LP** is a **continuous maximization** problem.
- **Feasible region** is defined by the **intersection** of all three **constraints**.
- **Optimal solution belongs** to the **feasible region** (**satisfies all constraints**) and **provides the highest** (**maximum**) **objective function** value.

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Example:

Consider that needs to invest $100 million. This sum has to be split between three types of investments: A, B, and C. We expect an annual return of 10%, 7%, and 3% for A, B, and C investments, respectively. Since investment A are more risky than B, the investment rules require the amount invested in A to be no more than one-third of the moneys invested in B. In addition, at least 25% of the total amount invested in A and B must be invested in C. How should we invest the money to maximize the return?

Decision variables

- x: amount invested in A
- y: amount invested in B
- z: amount invested in C

Linear program

maximize $0.10x + 0.07y + 0.03z$

subject to

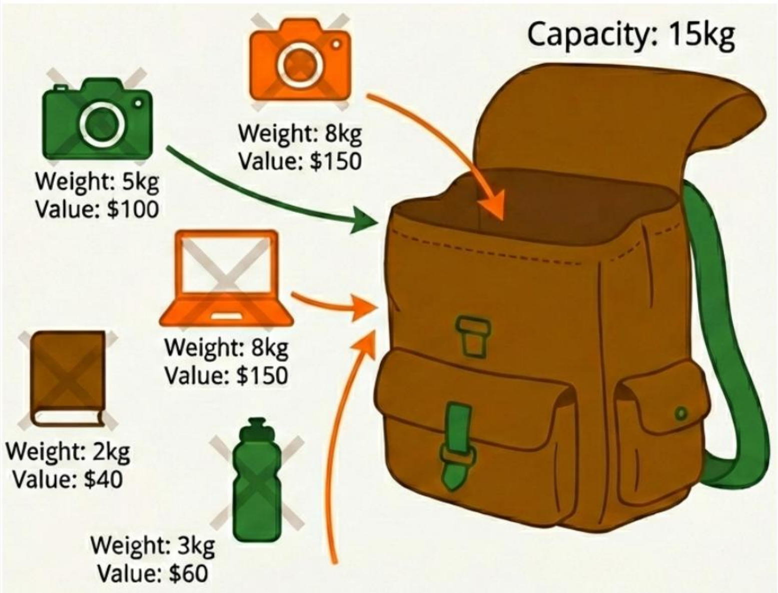
$$x + y + z = 100$$
$$x \le \frac{1}{3} y$$
$$z \ge 0.25(x + y)$$
$$x \ge 0,\ y \ge 0,\ z \ge 0$$

Optimal Solution

X=20; y=60; z=20

Objective function: 6.8

- The **Knapsack Problem (KP)** is a classic combinatorial optimization problem that can be described as follows: given a set of n objects, each with a weight w_i and a profit p_i, determine the subset of objects to include in a knapsack of capacity C such that the total profit is *maximized* without exceeding the knapsack's weight limit. Each object can either be included in the knapsack or not, leading to a binary decision for each object.



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Decision variables: Binary variables $x_i \in \{0,1\}, \forall i \in \{1,2,\ldots,n\}$ indicating whether object i is included in the knapsack ($x_i = 1$) or not ($x_i = 0$).

Objective function: Maximize the total profit of the selected objects in the knapsack.

$MaxZ = \sum_{i=1}^{n} p_i x_i$

Constraints: The total weight of the selected objects must not exceed the knapsack capacity C.

$\sum_{i=1}^{n} w_i x_i \leq C$

General formulation of the 0-1 Knapsack Problem

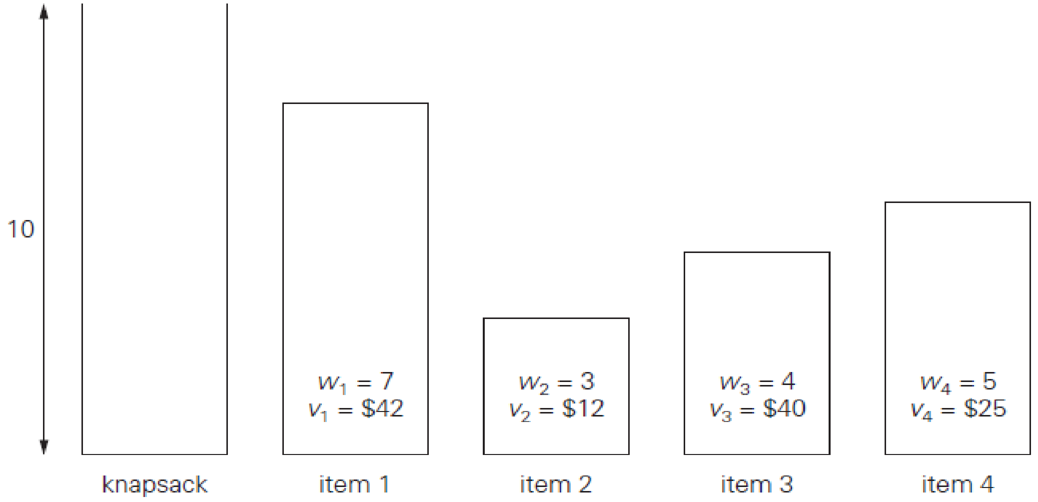
$$\text{Max } Z = \sum_{i=1}^{n} p_i x_i$$

$$\text{(s.t)} \begin{cases} \sum_{i=1}^{n} w_i x_i \leq C \\ \forall i \in \{1,2,\ldots,n\}, x_i \in \{0,1\} \end{cases}$$

Model Analysis

The model is a **Binary Integer Linear Programming (BILP)** with n **variables** and **1 constraint** (plus the binary constraints).

Example: Reducing the Knapsack problem to a linear problem



Linear program

maximize $42x_1 + 12x_2 + 40x_3 + 25x_4$

subject to

$$7x_1 + 3x_2 + 4x_3 + 5x_4 \leq 10$$

$$x_1, x_2, x_3, x_4 \in \{0,1\}$$

<u>Optimal Solution</u>

$x_1 = 0$

$x_2 = 0$

$x_3 = 1$

$x_4 = 1$

Objective function: 65

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Problem reduction –Linear Programming

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Example:Choclate manufacturer

Consider a chocolate manufacturing company that produces only two types of chocolate – A and B. Both the chocolates require Milk and Choco only. To manufacture each unit of A and B, the following quantities are required:

- Each unit of A requires 1 unit of Milk and 3 units of Choco
- Each unit of B requires 1 unit of Milk and 2 units of Choco

The company kitchen has a total of 5 units of Milk and 12 units of Choco. On each sale, the company makes a profit of:

- 6 sar per unit A sold
- 5 sar per unit B sold

Now, the company wishes to maximize its profit. How many units of A and B should it produce respectively?

Decision variables

The total number of units produced by A be = X

The total number of units produced by B be = Y

Linear program

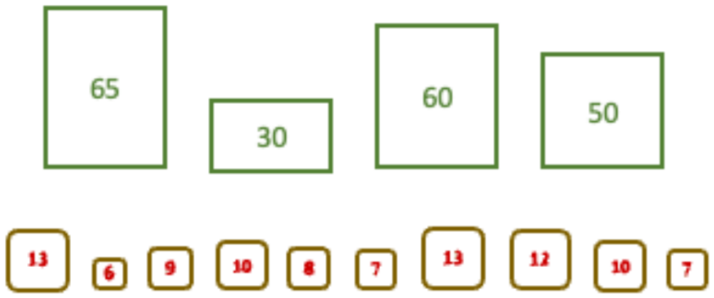
maximize $6X + 5Y$

subject to $X + Y \leq 5$

$3X + 2Y \leq 12$

$X \geq 0$ & $Y \geq 0$

Example: Bin Packing Problem



10 items 4 bins

The **Bin Packing Problem (BPP)** is a classic combinatorial optimization problem that can be described as follows: given a set of n items, each with a volume v_i, and a set of m bins, each with a maximum capacity c_i, determine the minimum number of bins required to pack all items such that the total volume of items in each bin does not exceed its capacity. Each item must be placed in exactly one bin. The tables beside illustrate an example of the Bin Packing Problem with $n = 10$ items and $m = 4$ bins.

Bin	Capacity
1	65
2	30
3	60
4	50

Item	Volume
1	13
2	6
3	9
4	10
5	8
6	7
7	13
8	12
9	10
10	7

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Specific formulation of the Bin Packing Problem

Decision variables:

- Binary variables $x_{ij} \in \{0,1\}, \forall i \in \{1,2,\dots,10\}, \forall j \in \{1,2,3,4\}$ indicating whether item i is placed in bin j ($x_{ij} = 1$) or not ($x_{ij} = 0$).
- Binary variables $y_j \in \{0,1\}, \forall j \in \{1,2,3,4\}$ indicating whether bin j is used ($y_j = 1$) or not ($y_j = 0$).

Objective function: Minimize the total number of bins used to pack all items.

$MinZ = y_1 + y_2 + y_3 + y_4 = \sum_{j=1}^{4} y_j$.

Constraints:

1. Each item must be placed in exactly one bin. $\sum_{j=1}^{4} x_{ij} = 1, \forall i \in \{1,2,\dots,10\}$
   - E.g., item 1: $x_{11} + x_{12} + x_{13} + x_{14} = 1$
   - E.g., item 2: $x_{21} + x_{22} + x_{23} + x_{24} = 1$
   - ...
   - E.g., item 10: $x_{10,1} + x_{10,2} + x_{10,3} + x_{10,4} = 1$
2. The total volume of items in each bin must not exceed its capacity. $\sum_{i=1}^{10} v_i x_{ij} \le c_j y_j, \forall j \in \{1,2,3,4\}$
   - E.g., bin 1: $13x_{11} + 6x_{21} + 9x_{31} + 10x_{41} + 8x_{51} + 7x_{61} + 13x_{71} + 12x_{81} + 10x_{91} + 7x_{10,1} \le 65y_1$
   - E.g., bin 2: $13x_{12} + 6x_{22} + 9x_{32} + 10x_{42} + 8x_{52} + 7x_{62} + 13x_{72} + 12x_{82} + 10x_{92} + 7x_{10,2} \le 30y_2$
   - E.g., bin 3: $13x_{13} + 6x_{23} + 9x_{33} + 10x_{43} + 8x_{53} + 7x_{63} + 13x_{73} + 12x_{83} + 10x_{93} + 7x_{10,3} \le 60y_3$
   - E.g., bin 4: $13x_{14} + 6x_{24} + 9x_{34} + 10x_{44} + 8x_{54} + 7x_{64} + 13x_{74} + 12x_{84} + 10x_{94} + 7x_{10,4} \le 50y_4$

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- The size of search space is $2^{n \times m}$, where n is the number of items and m is the number of bins.
  - The Bin Packing Problem is encountered in various real-world applications, such as resource allocation, logistics, and manufacturing.
  - *E*numerating all possible arrangements of items in bins to find the optimal solution is computationally infeasible for large n and m.
  - The table shows the combinatorial explosion of the search space as the number of items and bins increases (*consider that each possibility is evaluated in* 10^{-6} *seconds*):

Number of items (n)	Number of bins (m)	Approx. Time to enumerate (seconds)
5	3	34.9
10	4	$1.0995 \times 10^{12} = 34,844$ years
15	5	$3.2768 \times 10^{30} = 1.04 \times 10^{23}$ years
20	6	$9.2234 \times 10^{48} = 2.92 \times 10^{41}$ years